

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety and Tolerability of BMS-986408 Alone and in Combination With Nivolumab or Nivolumab and Ipilimumab in Participants With Advanced Solid Tumors **Short Title/Acronym:** BMS-986408 Monotherapy and Combination in Advanced Solid Tumors **Protocol Number:** CA099-003 (Secondary ID: NCI-2022-06372) **NCT Number:** NCT05407675 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** BMS-986408, Nivolumab, Ipilimumab, Rabeprazole (Trade Names: Aciphex, Pariet; ATC Code: A02BC04) **Preferred UMLS Name:** Antineoplastic Agents **Phase of Development:** Phase I/II **Medical Monitor:** Bristol-Myers Squibb Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety profile of BMS-986408 as monotherapy and in combination with nivolumab or nivolumab and ipilimumab to establish the maximum tolerated dose (MTD). **Secondary Objectives:** To determine the Recommended Phase 2 Dose (RP2D) that optimizes the pharmacokinetic/pharmacodynamic (PK/PD) relationship of BMS-986408; to evaluate PK parameters (Cmax, Tmax, AUC), Objective Response Rate (ORR), and Duration of Response (DOR).

2.2 Background & Rationale To address the critical unmet medical need for patients with advanced, unresectable, or metastatic solid tumors who have been refractory to, ineligible for, or intolerant of existing standard therapies. This study investigates whether the addition of BMS-986408 (a targeted DGKα inhibitor) can overcome immune hypofunctionality and enhance the efficacy of established immune checkpoint inhibitors like nivolumab and ipilimumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation and safety expansion cohorts) **Blinding:** Open-label **Randomization:** N/A (Cohort-based dose assignment)

3.2 Planned Enrollment & Duration Trial Status: COMPLETED **Target N\$:** 68 **Number of Sites:** Global, including multiple academic oncology centers (e.g., Dana-Farber Cancer Institute, Switzerland sites like Basel, Geneva, St. Gallen) **Estimated Study Start:** June 2022 **Estimated Primary Completion:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with a histologically or cytologically confirmed, advanced, unresectable/metastatic, solid malignancy of any histology measurable by Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. 2. Participants who have received, been refractory to, ineligible for, or intolerant of existing therapy(ies) known to provide clinical benefit for the condition of the participant. 3. Participants with melanoma should have documentation of mutation status for B-type Raf proto-oncogene (BRAF) and neuroblastoma ras viral oncogene homolog (NRAS). 4. Participants must have experienced radiographically documented progressive disease on or after the most recent therapy.

4.2 Exclusion Criteria 1. An active, known or suspected autoimmune disease. 2. Conditions requiring systemic treatment with either corticosteroids within 14 days or other immunosuppressive medications within 30 days of the first dose of study treatment. 3. Current or recent gastrointestinal disease or gastrointestinal surgery that could impact the absorption of the study drug. 4. Untreated central nervous system (CNS) metastases or leptomeningeal metastasis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ascending doses of BMS-986408 administered alone, or combined with standard dosing of Nivolumab or Nivolumab + Ipilimumab until MTD and RP2D are established. **Route of Administration:** Oral (BMS-986408) and Intravenous (Nivolumab/Ipilimumab). **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or consent withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive oncology care and palliative treatments not interacting with investigational pathways. **Prohibited:** Systemic corticosteroids (within 14 days) and immunosuppressive medications (within 30 days of the first dose).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans (RECIST v1.1), Labs
Baseline	Day 0	First Dose of BMS-986408 +/- Immunotherapy, Safety & PK Baseline
Interim	Cycle-dependent	Safety Labs for Dose-Limiting Toxicities (DLTs), PK/PD blood draws, Efficacy Assessments
End of Study	Progression	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Establishment of the Maximum Tolerated Dose (MTD) assessed via: 1. Number of Participants With Dose Limiting Toxicities (DLTs). 2. Number of Participants With Adverse Events (AEs). 3. Number of Participants Who Died.

Measurement Method: Clinical evaluation and safety laboratory analysis graded by CTCAE criteria.

7.2 Secondary & Exploratory Endpoints - Maximum Observed Plasma Concentration (C_{max}) of BMS-986408. - Time to Maximum Observed Plasma Concentration (T_{max}) of BMS-986408. - Area Under the Plasma Concentration-time Curve From Time 0 to Time of Last Quantifiable Concentration [AUC(0-T)]. - Determination of the Recommended Phase 2 Dose (RP2D). - Efficacy: Objective Response Rate (ORR) and Duration of Response (DOR) assessed by RECIST v1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by HCPs for DLTs and general toxicity associated with novel immunotherapies.

Serious Adverse Events (SAEs): Reported to Sponsor within standard 24-hour oncology protocols. **Data Monitoring Committee (DMC):** Internal/External safety review cohorts for dose-escalation clearance.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all treated participants) for DLT/AE determination; Efficacy Set for ORR and DOR endpoints.

Sample Size Justification: Based on standard Phase 1/2 dose-escalation statistical models required to identify MTD and RP2D.

Handling of Missing Data: Missing PK/PD data excluded from concentration curves; survival/response analyses censored at the date of last contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard oncology monitoring, integrating on-site checks and remote data verifications. **Audit Trail:** Mandatory eCRF locking and query resolution processes post-assessment.

11. Study Identifiers & Governance

Primary ID: CA099-003 **Posting ID:** 1014858546236547335 **Registry Number:** NCT05407675 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate the Safety and Tolerability of BMS-986408 Alone and in Combination With Nivolumab or Nivolumab and Ipilimumab in Participants With Advanced Solid Tumors **Official Regulatory Title:** A Phase 1/2 Study of BMS-986408 Alone and in Combination with Nivolumab or with Nivolumab and Ipilimumab in Participants with Advanced Solid Tumors

12. Clinical Framework (Oncology Specific adaptation)

Primary Condition/Disease Area: Advanced Solid Tumors (Name: Advanced Solid Malignancies) **Preferred Name:** Advanced Malignant Solid Neoplasm **Semantic Type:** Neoplastic Process **Definition:** A malignant solid neoplasm that has spread extensively to other anatomic sites or is no longer responding to treatment. **SNOMED ID:** 39958004 **SNOMED Hierarchy:** 785317006|Advanced solid malignant neoplasm|Advanced solid malignant neoplasm (disorder) **Medical Methodology:** Experimental targeted immunotherapy combination **Optimization Target:** Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Advanced/Refractory Solid Tumor Oncology Clinical Pathway **Education Protocol (HCP):** Dose-escalation rules, DLT management, and AE grading specific to immune-mediated toxicities **Education Protocol (Patient):** Informed consent detailing Phase 1/2 risks, visit cadence, and immunotherapy side effects **Quality Audit Mechanism:** Pharmacovigilance tracking and dose-escalation safety reviews

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression, death, or study withdrawal **Onsite Visit Cadence:** Per protocol treatment cycles (e.g., Day 1 of each cycle for dosing and safety labs) **Remote Monitoring Frequency:** Regular interval symptom checks for immune-mediated AEs **Checklist Review Interval:** Efficacy tumor imaging (CT/MRI) per RECIST v1.1 intervals

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Insert Investigator Name]				____		

[DD-MMM-YYYY] | | Clinical Operations Lead | [Insert COL Name] |
____ | [DD-MMM-YYYY] | | Lead Statistician | [Insert Statistician
Name]| _____ | [DD-MMM-YYYY] |